Background for quantum cellular automata

What is CA (cellular automata)

CA consists of an array of identical cells, each of which can be in one of a finite number of states. The cells update their states synchronously according to a local function G in discrete time steps.

What can CA (cellular automata) do?

It can evolve automatically according to the local rules without any external control.

Cellular Automata(CA)

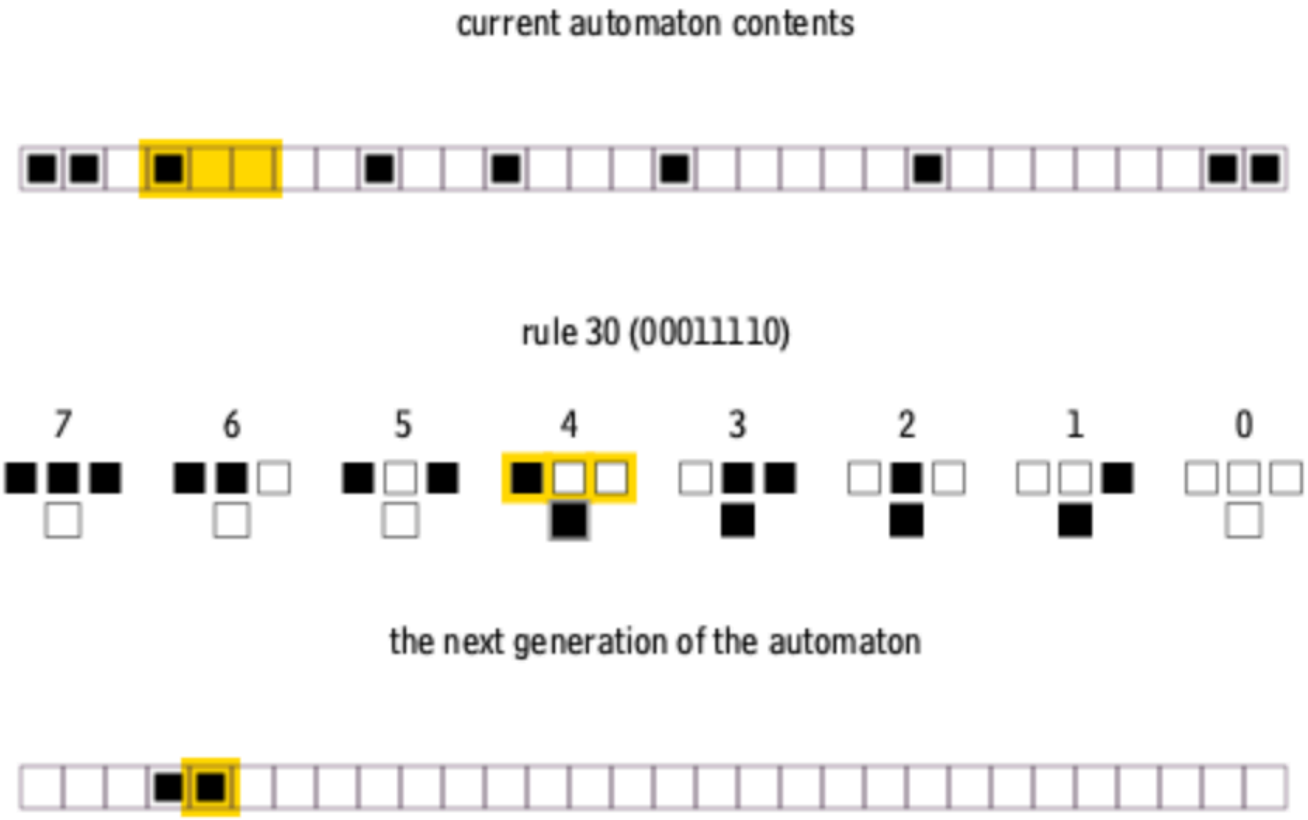

Figure 1: Cellular Automata

Cellular Automata(CA)

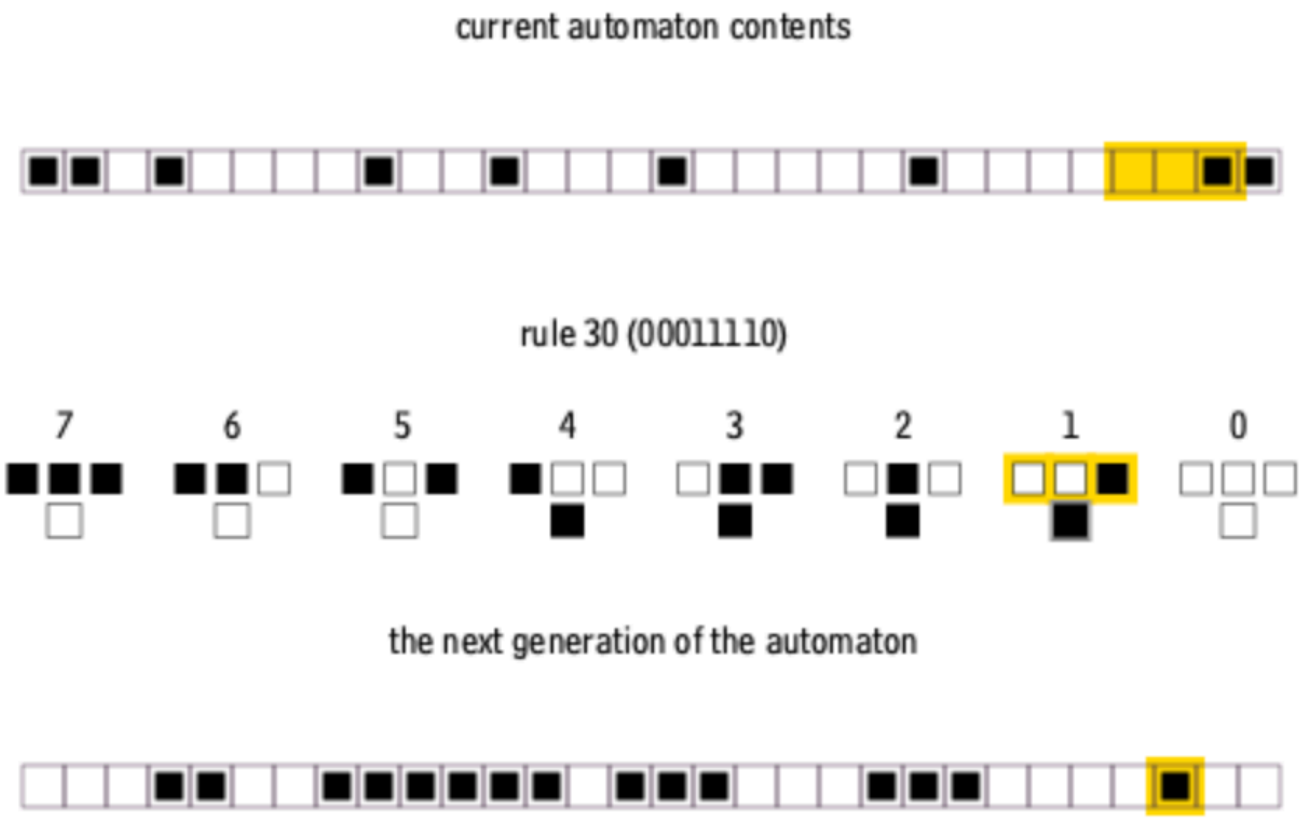

Figure 2: Cellular Automata

Cellular Automata(CA)

CA (cellular automata) for universal computation

Universal CA is universal for classical computation [1], which, likes Turing machine, exhibits universality.

Example: Game of Life （Universal for TM）

CA have been widely used in the study of complex systems, such as the Game of Life by Conway.

In 2D grid, each cell has 8 neighbors.

If the cell is alive, then it stays alive if it has either 2 or 3 live neighbors

If the cell is dead, then it springs to life only in the case that it has 3 live neighbors

Cellular Automata(CA)

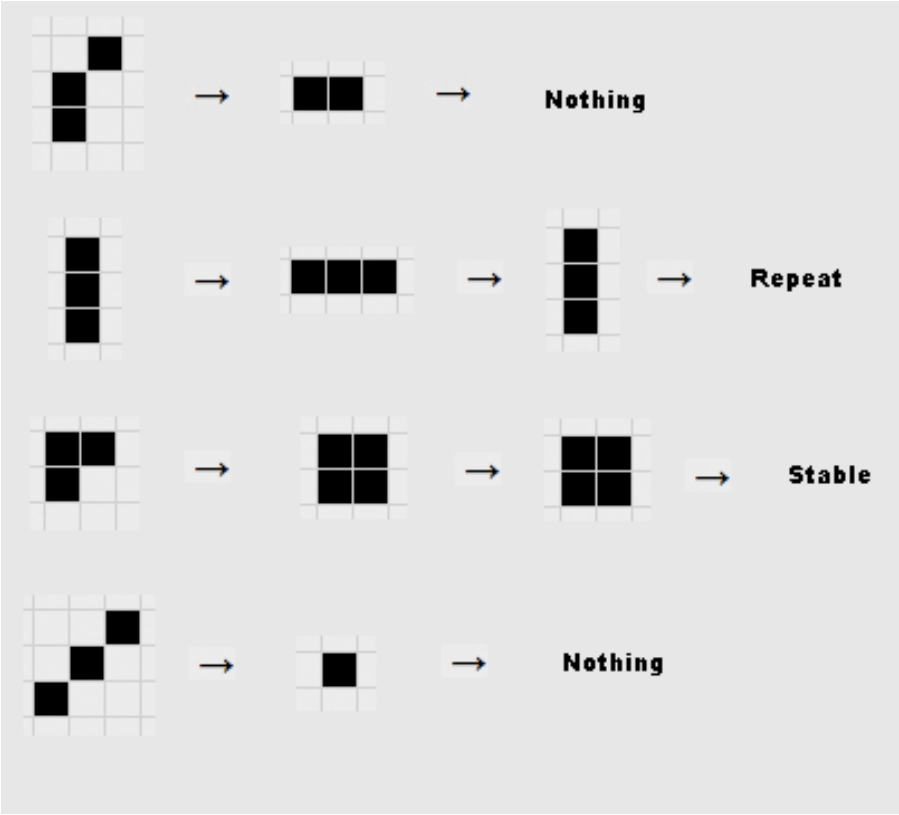

Figure 3: Game of Life by Conway

Introduction of QCA

Cellular Automata(CA)

QCA

QCA is a quantum version of CA, which can perform quantum computation.

- cells = qubits
- states = quantum states
- local function G = unitary operator
- no external control (Low noise)

Universality of QCA

The QCA-related results are focus on universality. Proving the universality of QCA in the sense of Quantum Turing Machine [2] is necessary.

- Proving that the evolution operator of QCA can cover all the unitary operators.
- Prove that QCA is equivalent to a known universal quantum gate set.

Quantum Cellular Automata

Main category

- Construction of intrinsically universal QCA.[3], [4]
- QCA for error correction. [5], [6]
- QCA based on Rydberg atom array[7], [8]
- Many-body simulation with QCA.[9],[10],[11]
- Numerical simulation of QCA (under noise).[12], [13]
- Entanglement dynamics in QCA.[14],[15]

Interesting works under QCA models

- Margolus partitioning: A general 2D lattice sites are partitioned into cells of size 2×2
- Separate partitioning for different time steps. (Overlap)

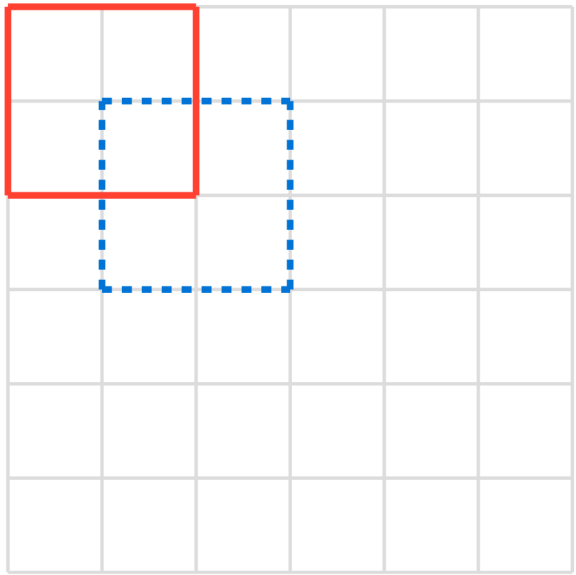

Figure 4: Margolus's physics-like cellular automata

Intrinsically universal QCA

Intrinsically universal 1D QCA

- The 1D intrinsically universal QCA is able to simulate all other 1D QCA[16]。
- Universal QCA(multi-steps)->simulated QCA(one-step)

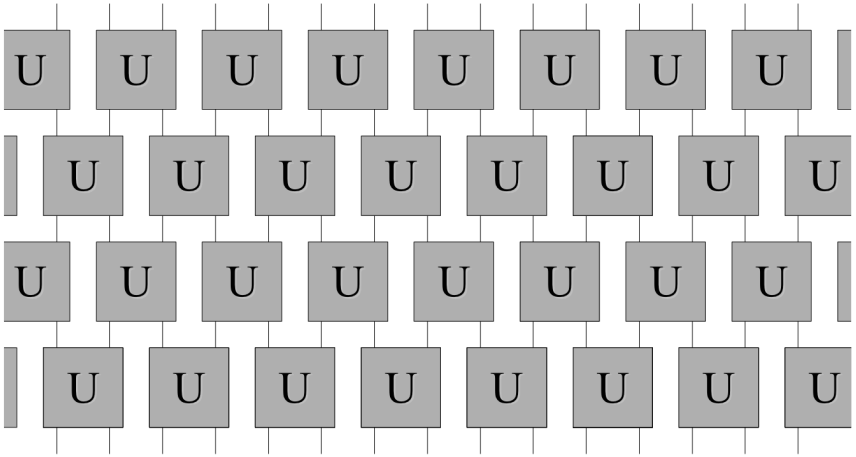

Figure 5: Partitioned one-dimensional QCA with scattering unitary U which gets applied upon two cells. Time flows upwards.

Intrinsically universal QCA

- Intrinsic simulation of a QCA by another:

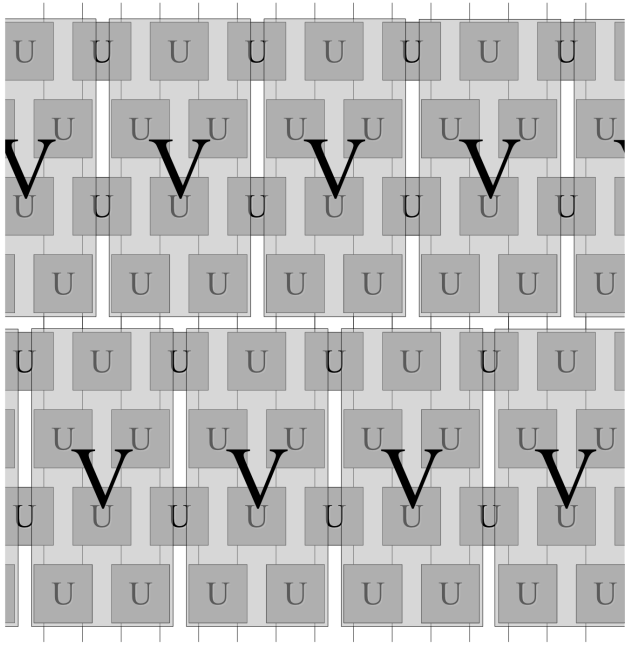

Figure 6: Intrinsic simulation of a QCA by another: The universal U-QCA and simulated V-QCA. In this case we need two cells of the U-QCA in order to encode one cell of the V-QCA, and we need to run the U-QCA for four time steps in order to simulate one time step of the V-QCA.

Intrinsically universal QCA

Intrinsic Simulation [4]

Definition 10 *(Intrinsic simulation).* Consider Σ_G and Σ_H, two alphabets with distinguished quiescent states q_G and q_H, and two QCA G and H over these alphabets. We say that G intrinsically simulates H if and only if there exists G', some grouping of G, and H', some grouping of H, such that G' directly simulates H'.

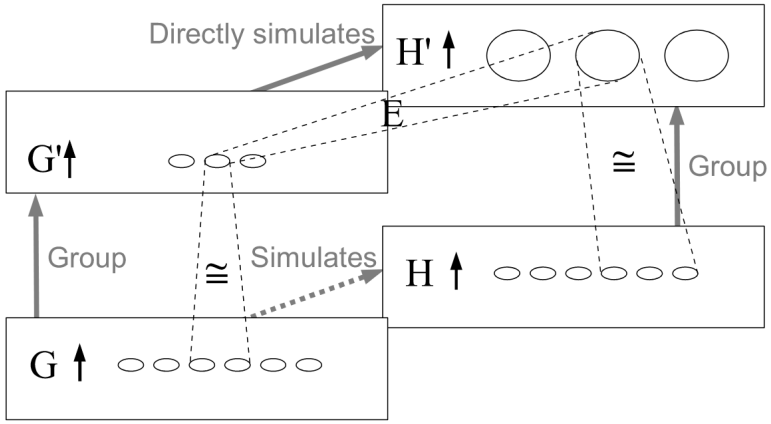

Figure 8: The concept of intrinsic simulation made formal.

Intrinsically universal QCA

They proposed nD PQCA which can simulate all other nD PQCA and have arbitrary precision[4]. The key is to determine the universal scattered unitary.

Partitioned QCA (PQCA)

The cells of QCA are devided into many non-overlapping blocks, and each block updates independently according to a specific rule. The blocks interact with each other in an alternating or staged manner, and the evolution rules are also processed in blocks. It is important to determine the universal scattered unitary.

Intrinsically universal QCA

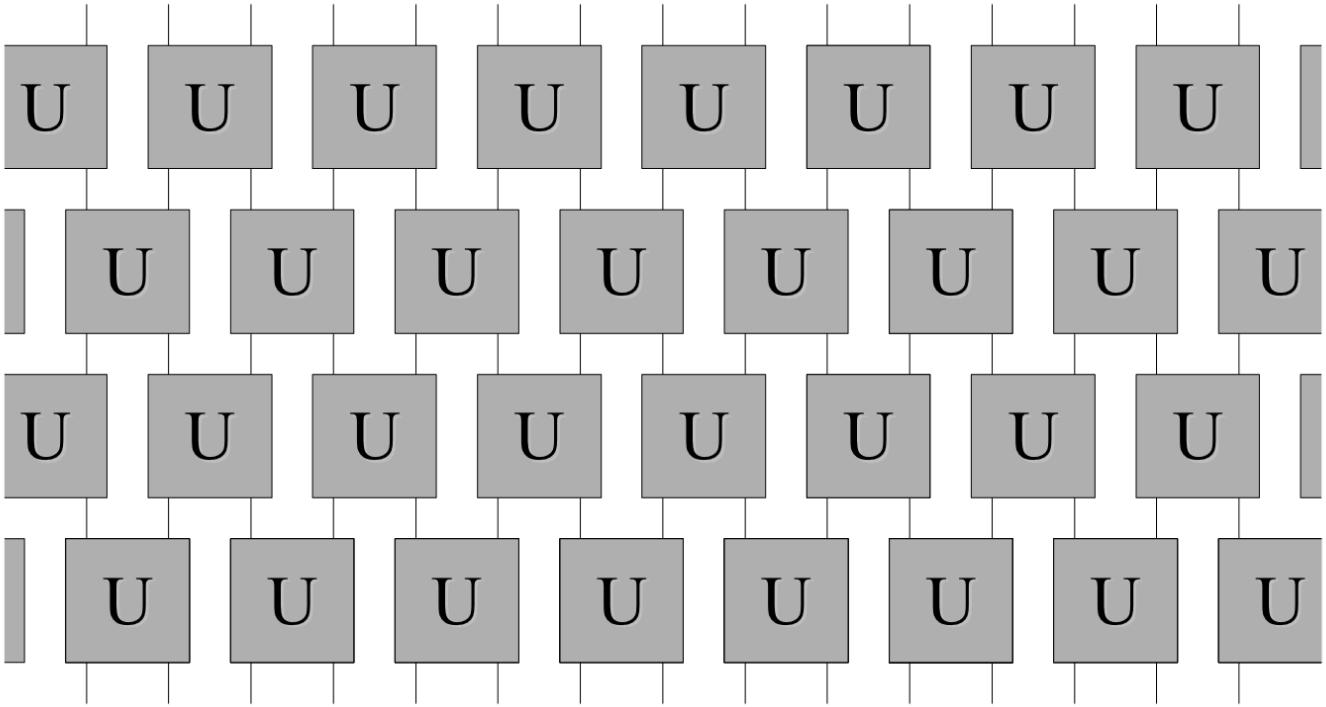

Figure 9: Partitioned one-dimensional QCA with scattering unitary U.

Intrinsically universal QCA

Block QCA (BQCA)

Quantisations of block representations of CA are generally presented as two-layer. We can assume that U_i is the unitary operator that acts on the ith block of cells. The evolution of the QCA is given by the following figure:

Intrinsically universal QCA

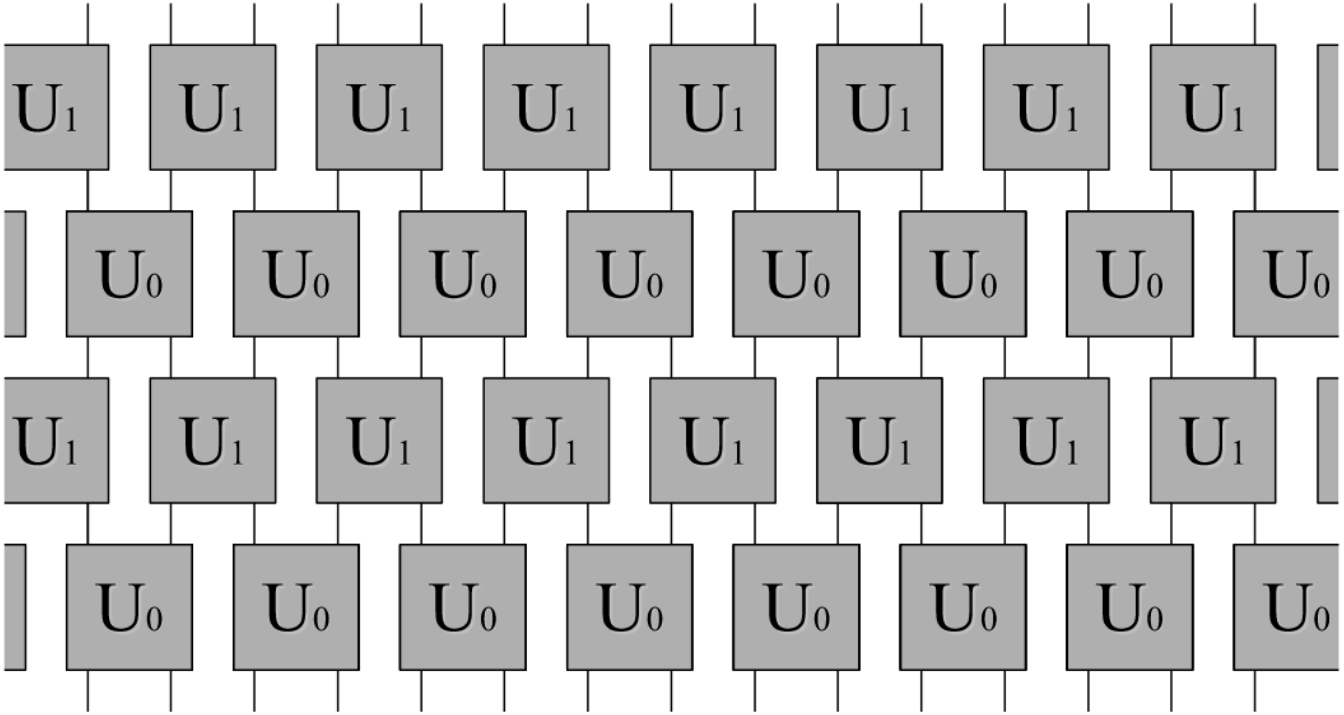

Figure 10: Two-layer block representation QCA

Intrinsically universal QCA

PQCA simulating BQCA

The difference is that two layers of BQCA are different. Thus, we need to use a U defined PQCA, with a U capable of performing U_0 and U_1.

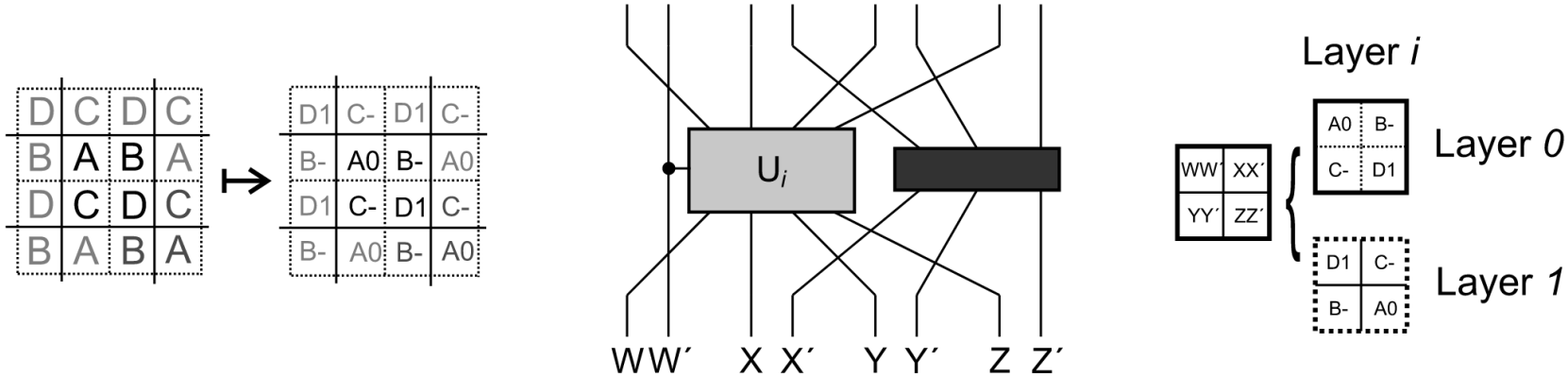

Figure 11: PQCA simulating a BQCA: The QCA is decorated with control qubits following a simple encoding procedure (left), which allow the scattering unitary U (centre) to act as either U0 or U1, according to the layer (right). The black box can be any unitary.

Intrinsically universal QCA

BQCA simulating QCA

If we can use PQCA to simulate BQCA, we can achieve the simulation of QCA by PQCA.

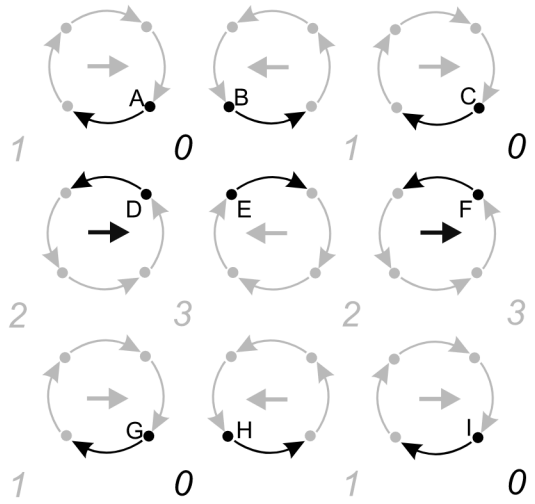

Figure 12: BQCA simulating QCA.

Intrinsically universal QCA

Quantum cellular automata are alternative quantum-computing paradigms to quantum Turing machines and quantum circuits. Their working mechanisms are inherently automated, therefore measurement free, and they act in a translation invariant manner on all cells or qudits of a register, generating a global rule that updates cell states locally, i.e., based solely on the states of their neighbors. Although desirable features in many applications, it is generally not clear to which extent these fully automated discrete-time local updates can generate and sustain long-range order in the (noisy) systems they act upon. In particular, whether and how quantum cellular automata can perform quantum error correction remain open questions. We close this conceptual gap by proposing quantum cellular automata with quantum-error-correction capabilities. We design and investigate two (quasi)one dimensional quantum cellular automata based on known classical cellular-automata rules with density-classification capabilities, namely the local majority voting and the two-line voting. We investigate the performances of those quantum cellular automata as quantum-memory components by simulating the number of update steps required for the logical information they act upon to be afflicted by a logical bit flip. The proposed designs pave a way to further explore the potential of new types of quantum cellular automata with built-in quantum-error-correction capabilities.

Figure 13: QCA for error correction

QCA for error correction

Error correction with CAs has been studied as a density-classification problem[6]

Density Classification problem

The DC problem is a generalization of the majority problem, which is to determine whether the majority of cells are in a certain state.

Local majority voting

If at least two of the three neighbors are 1, the updated state of the central cell is 1. Otherwise it is 0. For example, the evolution of 1D chain: [0 1 0 1 1 0 1 0 1]->[0 0 1 1 1 1 0 1 0]->[0 0 1 1 1 1 1 0 0]. If 1 is the majority in the initial state, it tends to be all 1 eventually. Conversely it goes to all zeros

QCA for error correction

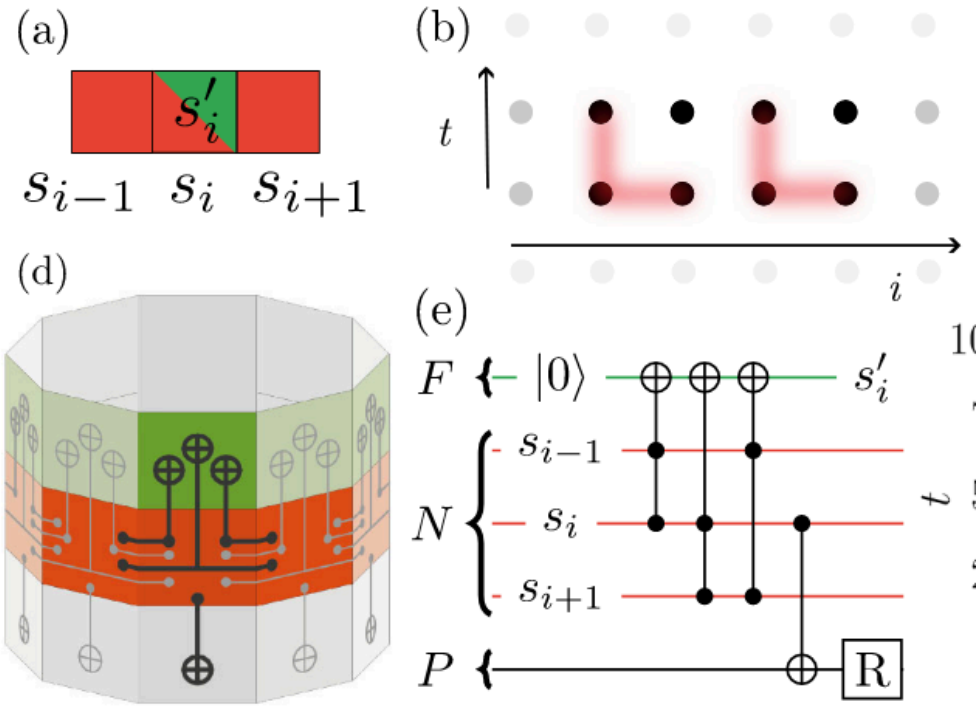

Figure 14: Local majority voting

QCA for error correction

Two-line voting (better performance under noise)

QTLV can solve the majority voting problem under noise. It is A 2D CA model with high robust. There are top line and bottom line. The updated state depends on the states of the neighbors in both top line and bottom line.

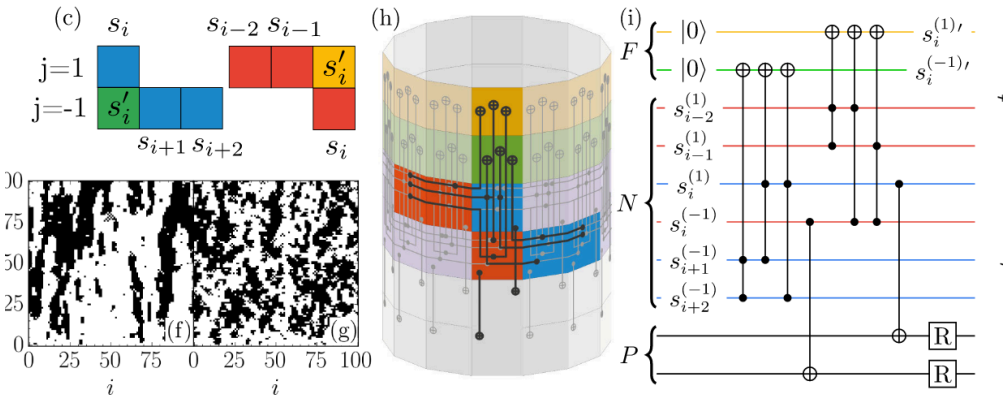

Figure 15: Two-line voting

QCA for error correction

1D topological QEC with symmetry-constrained CA [5]

- Focusing on the 1D Majorana chain
- constructing a strictly local decoder based on self-dual CA for error correction

Decoder: Scalability, with exponentially growing decoherence times in the presence of noise. (Essential for large-scale quantum storage).

QCA for error correction

Strictly local one-dimensional topological quantum error correction with symmetry-constrained cellular automata

Nicolai Lang⋆ and Hans Peter Büchler

Institute for Theoretical Physics III and Center for Integrated Quantum Science and Technology, University of Stuttgart, 70550 Stuttgart, Germany

⋆ nicolai@itp3.uni-stuttgart.de

Abstract

Active quantum error correction on topological codes is one of the most promising routes to long-term qubit storage. In view of future applications, the scalability of the used decoding algorithms in physical implementations is crucial. In this work, we focus on the one-dimensional Majorana chain and construct a strictly local decoder based on a self-dual cellular automaton. We study numerically and analytically its performance and exploit these results to contrive a scalable decoder with exponentially growing decoherence times in the presence of noise. Our results pave the way for scalable and modular designs of actively corrected one-dimensional topological quantum memories.

Figure 16: 1D topological QEC

QCA for error correction

Universal Quantum Computation in Globally Driven Rydberg Atom Arrays

Francesco Cesa[1,2,3,4,*] and Hannes Pichler[1,2,†]
[1]*Institute for Theoretical Physics, University of Innsbruck, Innsbruck A-6020, Austria*
[2]*Institute for Quantum Optics and Quantum Information, Austrian Academy of Sciences, Innsbruck A-6020, Austria*
[3]*Department of Physics, University of Trieste, Strada Costiera 11, 34151 Trieste, Italy*
[4]*Istituto Nazionale di Fisica Nucleare, Trieste Section, Via Valerio 2, 34127 Trieste, Italy*

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We develop a model for quantum computation with Rydberg atom arrays, which only relies on global driving, without the need of local addressing of the qubits: any circuit is executed by a sequence of global, resonant laser pulses on a static atomic arrangement. We present two constructions: for the first, the circuit is imprinted in the trap positions of the atoms and executed by the pulses; for the second, the atom arrangement is circuit-independent, and the algorithm is entirely encoded in the global driving sequence. Our results show in particular that a quadratic overhead in atom number is sufficient to eliminate the need for local control to realize a universal quantum processor. We give explicit protocols for all steps of an arbitrary quantum computation, and discuss strategies for error suppression specific to our model. Our scheme is based on dual-species processors with atoms subjected to Rydberg blockade constraints, but it might be transposed to other setups as well.

Figure 17: Hamiltonian quantum cellular automata

QCA based on Rydberg atom arrays

- global driving, without local addressing [7]
- two different structures are proposed:
  - imprint the circuit onto the trap positions of the atoms and then apply pulses.
  - arrange the atoms in a circuit-independent way, with the algorithm completely encoded in the global driving sequence.

QCA based on Rydberg atom arrays

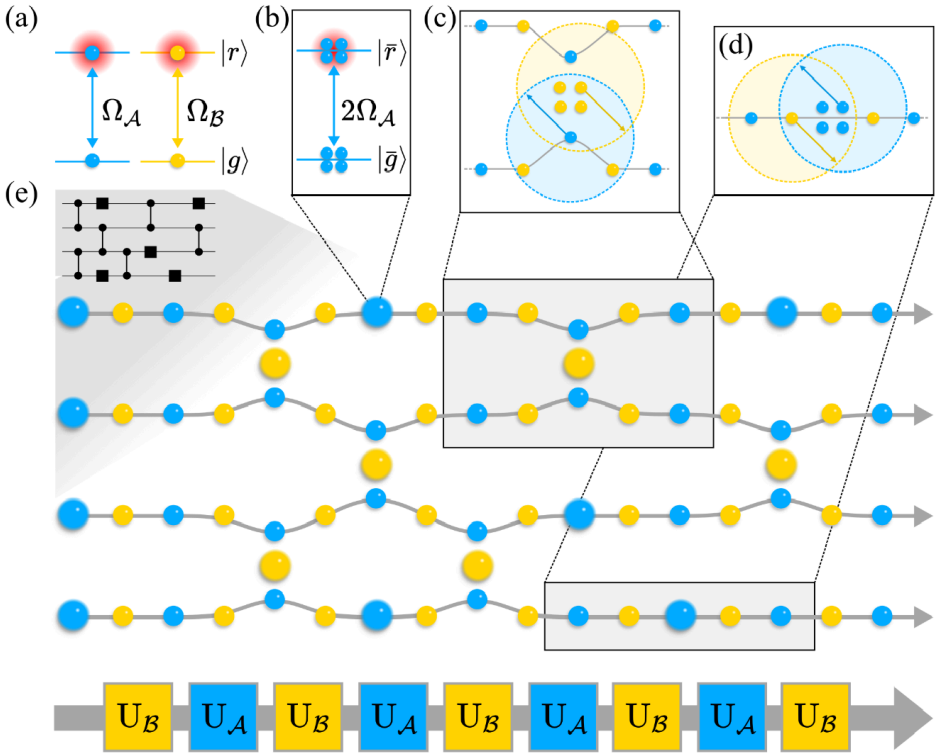

Figure 18: Atomic arrangement

QCA based on Rydberg atom arrays

1D wires of atoms

- qubit state($|\Psi(k)\rangle$) is located at a single atom k.
- After applying pulses:

$$|\Psi(k)\rangle \rightarrow |\Psi(k+1)\rangle$$

- The qubit propagates through the wire as the pulses progress - defining an information flow.

QCA based on Rydberg atom arrays

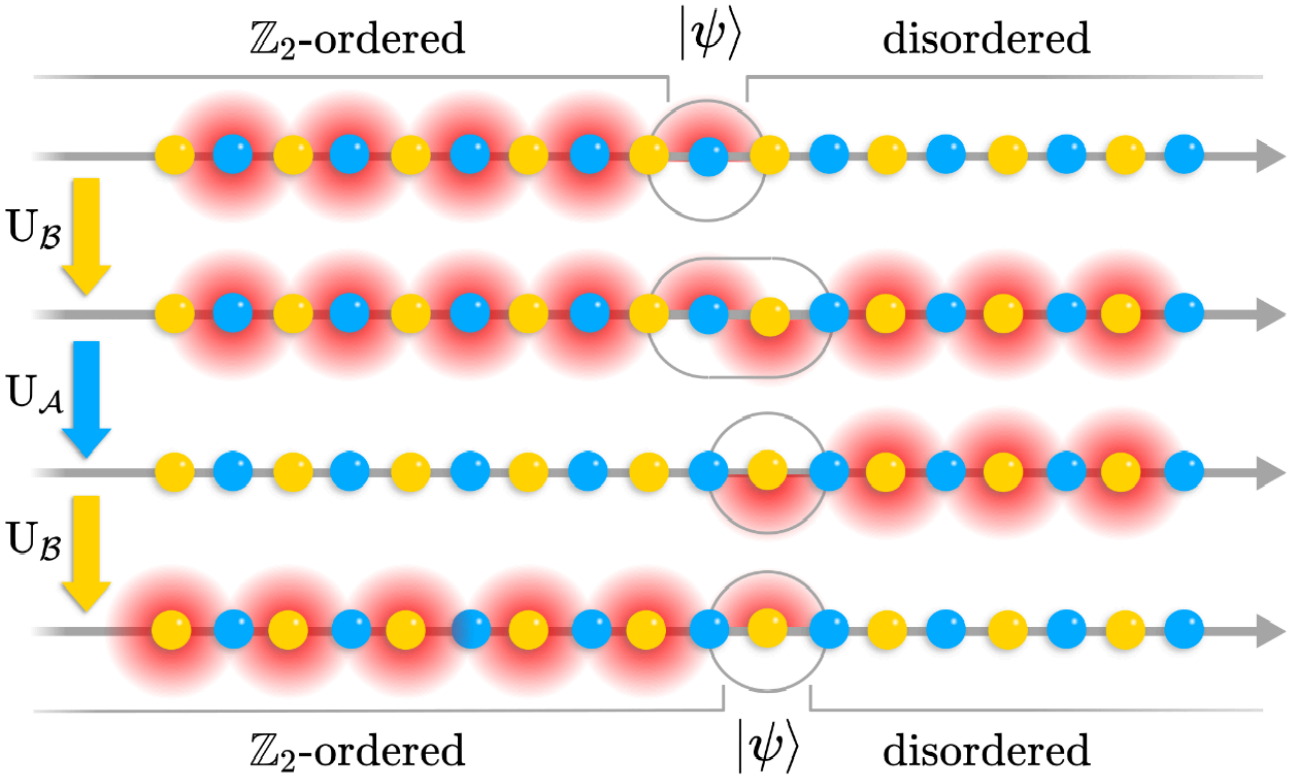

Figure 19: state transported through a wire via global pulses.

QCA based on Rydberg atom arrays

Unitary and Nonunitary Quantum Cellular Automata with Rydberg Arrays

T. M. Wintermantel,[1,2] Y. Wang,[2] G. Lochead,[2] S. Shevate,[2] G. K. Brennen,[3] and S. Whitlock[2,*]
[1]*Physikalisches Institut, Universität Heidelberg, Im Neuenheimer Feld 226, 69120 Heidelberg, Germany*
[2]*ISIS (UMR 7006) and IPCMS (UMR 7504), University of Strasbourg and CNRS, 67000 Strasbourg, France*
[3]*Center for Engineered Quantum Systems, Department of Physics & Astronomy, Macquarie University, 2109 New South Wales, Australia*

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We propose a physical realization of quantum cellular automata (QCA) using arrays of ultracold atoms excited to Rydberg states. The key ingredient is the use of programmable multifrequency couplings which generalize the Rydberg blockade and facilitation effects to a broader set of nonadditive, unitary and nonunitary (dissipative) conditional interactions. Focusing on a 1D array we define a set of elementary QCA rules that generate complex and varied quantum dynamical behavior. Finally, we demonstrate theoretically that Rydberg QCA is ideally suited for variational quantum optimization protocols and quantum state engineering by finding parameters that generate highly entangled states as the steady state of the quantum dynamics.

QCA based on Rydberg atom arrays

- Unitary (reversible) and non-Unitary (dissipative of intermediate state) QCA can be implemented with Rydberg atoms.[8]

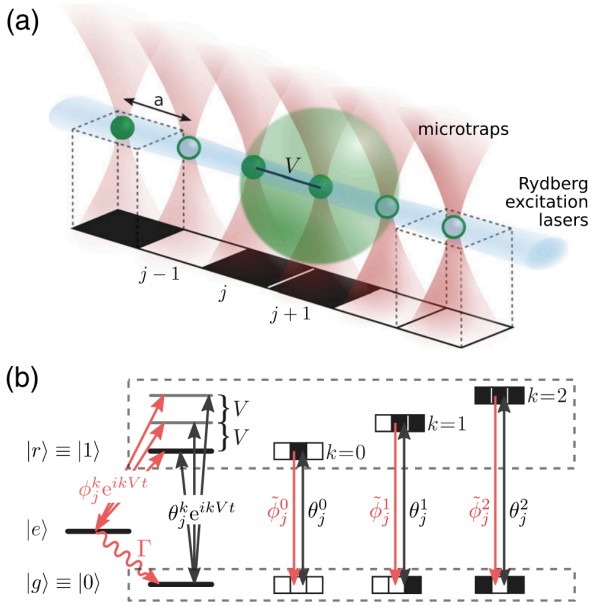

Figure 21: Rydberg QCA

QCA based on Rydberg atom arrays

Nonequilibrium Phase Transitions in (1 + 1)-Dimensional Quantum Cellular Automata with Controllable Quantum Correlations

Edward Gillman,[1,2] Federico Carollo,[3] and Igor Lesanovsky[1,2,3]

[1]*School of Physics and Astronomy, University of Nottingham, Nottingham, NG7 2RD, United Kingdom*

[2]*Centre for the Mathematics and Theoretical Physics of Quantum Non-Equilibrium Systems, University of Nottingham, Nottingham, NG7 2RD, United Kingdom*

[3]*Institut für Theoretische Physik, Universität Tübingen, Auf der Morgenstelle 14, 72076 Tübingen, Germany*

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Motivated by recent progress in the experimental development of quantum simulators based on Rydberg atoms, we introduce and investigate the dynamics of a class of (1 + 1)-dimensional quantum cellular automata. These nonequilibrium many-body models, which are quantum generalizations of the Domany-Kinzel cellular automaton, possess two key features: they display stationary behavior and nonequilibrium phase transitions despite being isolated systems. Moreover, they permit the controlled introduction of local quantum correlations, which allows for the impact of quantumness on the dynamics and phase transition to be assessed. We show that projected entangled pair state tensor networks permit a natural and efficient representation of the cellular automaton. Here, the degree of quantumness and complexity of the dynamics is reflected in the difficulty of contracting the tensor network.

Many-body simulation with QCA.

Nonequilibrium Phase Transition in (1+1)D QCA

- (1+1)D QCA, dimension of time and space.
- Based on Domany-Kinzel cellular automaton
  - stationary behavior and nonequilibrium phase transitions despite being isolated systems
  - permitting the controlled introduction of local quantum correlations.
- projected entangled pair state (PEPS) tensor networks for natural and efficient representation of the cellular automaton.[9]

Many-body simulation with QCA.

Topological Phases of Many-Body Localized Systems: Beyond Eigenstate Order

David M. Long[1,*] and Dominic V. Else[2]

[1] *Condensed Matter Theory Center and Joint Quantum Institute, Department of Physics, The University of Maryland, College Park, Maryland 20742, USA*

[2] *Perimeter Institute for Theoretical Physics, Waterloo, Ontario N2L 2Y5, Canada*

(Dated: August 23, 2024)

Many-body localization (MBL) lends remarkable robustness to nonequilibrium phases of matter. Such phases can show topological and symmetry breaking order in their ground and excited states, but they may also belong to an *anomalous localized topological phase* (ALT phase). All eigenstates in an ALT phase are trivial, in that they can be deformed to product states, but the entire Hamiltonian cannot be deformed to a trivial localized model without going through a delocalization transition. Using a correspondence between MBL phases with short-ranged entanglement and locality preserving unitaries—called quantum cellular automata (QCA)—we reduce the classification of ALT phases to that of QCA. This method extends to periodically (Floquet) and quasiperiodically driven ALT phases, and captures anomalous Floquet phases within the same framework as static phases. The QCA framework further generalizes to include symmetry-enriched ALT phases (SALT phases), and provides a large class of soluble models suitable for realization in quantum simulators. In systematizing the study of ALT phases, we both greatly extend the classification of interacting nonequilibrium systems and clarify a confusion in the literature which implicitly equates nontrivial Hamiltonians with nontrivial ground states.

Many-body simulation with QCA.

Topological Phases of Many-Body Localized Systems: Beyond Eigenstate Order

- Many-body localization system, for instance, Anomalous localized topological phase, there eigenstates are trivial but the entire Hamiltonian cannot be deformed to a trivial localized model without going through a delocalization transition. [11]
- Method: mapping complex MBL systems to QCA, and classifying ALTs by QCA rules.

Many-body simulation with QCA.

This work expands a previous block-partitioned quantum cellular automata (BQCA) model proposed by Brennen and Williams [Phys. Rev. A. **68**, 042311 (2003)] to incorporate physically realistic error models. These include timing errors in the form of over- and under-rotations of quantum states during computational gate sequences, stochastic phase and bit flip errors, as well as undesired two-bit interactions occurring during single-bit gate portions of an update sequence. A compensation method to counteract the undesired pairwise interactions is proposed and investigated. Each of these error models is implemented using Monte Carlo simulations for stochastic errors and modifications to the prescribed gate sequences to account for coherent over-rotations. The impact of these various errors on the function of a QCA gate sequence is evaluated using the fidelity of the final state calculated for four quantum information processing protocols of interest: state transfer, state swap, GHZ state generation, and entangled pair generation. © *2016 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (http://creativecommons.org/licenses/by/4.0/).* [http://dx.doi.org/10.1063/1.4963300]

Numerical simulation of QCA (under noise).

Performance of 1D quantum cellular automata in the presence of error

- Based on Block QCA model, they consider realistic physical error including over- and under- rotations of quantum states during gate sequence, stochastic phase and bit flip errors, and undesired interactions with neighbors when carrying out single qubit operation (which can be eliminated through compensation method). [12]

Numerical simulation of QCA (under noise).

Abstract

Cellular automata are interacting classical bits that display diverse emergent behaviors, from fractals to random-number generators to Turing-complete computation. We discover that quantum cellular automata (QCA) can exhibit complexity in the sense of the complexity science that describes biology, sociology, and economics. QCA exhibit complexity when evolving under 'Goldilocks rules' that we define by balancing activity and stasis. Our Goldilocks rules generate robust dynamical features (entangled breathers), network structure and dynamics consistent with complexity, and persistent entropy fluctuations. Present-day experimental platforms—Rydberg arrays, trapped ions, and superconducting qubits—can implement our Goldilocks protocols, making testable the link between complexity science and quantum computation exposed by our QCA.

Entangled quantum cellular automata, physical complexity, and Goldilocks rules

Numerical simulation of QCA (under noise).

- Goldilocks Rule: The quantum rule is in intermediate state (not too active nor static) and result in complex behavior. [13]

Numerical simulation of QCA (under noise).

Entanglement dynamics and ergodicity breaking in a quantum cellular automaton

Kevissen Sellapillay* and Alberto D. Verga†
Aix-Marseille Université, Université de Toulon, CNRS, CPT, 13009 Marseille, France

Giuseppe Di Molfetta‡
Aix-Marseille Université, LIS, Campus de Luminy, 13288 Marseille, France

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Ergodicity breaking is observed in the blockade regime of Rydberg atom arrays, in the form of low entanglement eigenstates known as scars, which fail to thermalize. The signature of these states persists in periodically driven systems, where they coexist with an extensive number of chaotic states. Here we investigate a quantum cellular automaton based on the classical rule that updates a site if its two neighbors are in the lower state. We show that the breaking of ergodicity extends to chaotic states. The dynamical breaking of ergodicity is controlled by chiral quasiparticle excitations which propagate entanglement. Evidence of nonlocal entanglement is found, showing that these nonthermal chaotic states may be useful to quantum computation.

Entanglement dynamics in QCA.

Entanglement dynamics and ergodicity breaking in QCA

- QCA: classical rule that updates a site if its two neighbors are in the lower state.
- showing that the breaking of ergodicity extends to chaotic states.
- Reason of breaking of ergodicity: chiral quasiparticle excitations which propagate entanglement.[10]

Entanglement dynamics in QCA.

Entanglement dynamics in one-dimensional quantum cellular automata

Gavin K. Brennen and Jamie E. Williams
National Institute of Standards and Technology, Gaithersburg, Maryland 20899-8423, USA
(Received 6 June 2003; published 13 October 2003)

Several proposed schemes for the physical realization of a quantum computer consist of qubits arranged in a cellular array. In the quantum circuit model of quantum computation, an often complex series of two-qubit gate operations is required between arbitrarily distant pairs of lattice qubits. An alternative model of quantum computation based on quantum cellular automata (QCA) requires only homogeneous local interactions that can be implemented in parallel. This would be a huge simplification in an actual experiment. We find some minimal physical requirements for the construction of unitary QCA in a one-dimensional Ising spin chain and demonstrate optimal pulse sequences for information transport and entanglement distribution. We also introduce the theory of nonunitary QCA and show by example that nonunitary rules can generate environment assisted entanglement.

Entanglement dynamics in QCA.

Entanglement dynamics in one-dimensional quantum cellular automata

- In general quantum computation, complex operation on certain qubits are required.
- Quantum computation based on QCA requires only homogeneous local interactions that can be implemented in parallel.
- In 1D Ising model, they show some minimal physical requirements for constructing QCA. And demonstrate optimal pulse sequence for information transport and entanglement distribution.
- Some non-Unitary operation (environment disturbed situation) can generate environment assisted entanglement.[14]

Entanglement dynamics in QCA.

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